

Speech Reduction in French: The Relationship Between Vowel Space and Articulation Dynamics

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Abstract

Reduction is an inherent characteristic of conversations, reflecting the dynamic adaptability of language. This study examined the link between vowel space and non-lexicalized reductions (NLR) in spontaneous French speech. The hypothesis posited that speakers with smaller vowel spaces —indicating centralized, less distinct vowels— would produce more NLR, defined as temporally compressed speech zones. Results showed that smaller vowel space (pVSA) predicted greater NLR only when articulation rate was considered, highlighting an interaction between spatial and temporal speech dynamics. Articulation rate was the strongest predictor, supporting theories that link faster speech to reduced articulatory precision. Vowel Distinctiveness Index (VDI) did not significantly predict NLR, suggesting it reflects broader systemic patterns rather than local reductions. These findings emphasized the interplay of temporal dynamics and individual articulation strategies in shaping reduction in speech.

Index Terms: speech reduction, vowel space area, vowel distinctiveness, spontaneous speech

1. Introduction

Spoken language is inherently shaped by variability in the speech signal, which arises from the continuous adaptation of speech production to contextual demands [1]. A key manifestation of this variability is speech reduction, where articulatory effort is minimized, often leading to shorter durations and less precise articulation. This process, particularly prominent in conversational speech, affects vowels significantly by altering their spectral and temporal characteristics [2].

Reduction can affect phonemes, syllables, and even entire words, resulting in diminished acoustic and phonetic substance [3] while still allowing a smooth interaction between speakers and listeners. Reduction often manifests as vowel centralization, resulting in smaller vowel space areas and less distinct vowel categories [4; 5]. Such centralization arises from formant undershoot during shorter vowel durations, which limits articulators from reaching vowel-specific acoustic targets. These changes can be observed as compressed F1 x F2 spaces, with deviations from prototypical vowel qualities [6; 7]. Factors such as speaking style, prosody, word frequency, and speech rate amplify these effects [8; 9; 10], making vowel space reduction a multifaceted phenomenon.

In spontaneous speech, reduction is not limited to vowels but occurs across various linguistic levels. Therefore, we propose a distinction between lexicalized and non-lexicalized reductions. Lexicalized reductions, such as regular examples in French like “*je ne sais pas*” [ʒə nə sɛ pa] reduced to [ʃɛpa] or [ʃpa], and “*je suis*” [ʒə sɥi] often produced as [ʃɥi], have been

more extensively studied due to their consistency and reproducibility across speakers and contexts as they involve fixed and predictable forms. Examples in English include “*going to*” pronounced as [ɡʊnə] and “*don’t know*” produced as [dɒnoʊ].

However, a substantial number of reductions do not conform to specific lexical representations and are barely perceptible to listeners. These non-lexicalized reductions involve temporally compressed segments and lack stable or consistent forms across speakers and contexts. They are context-dependent, and significantly influenced by articulation rate, morphosyntactic structure, or prosodic patterns [11]. These reductions frequently obscure individual segments, compressing phonemes into shorter intervals while still enabling listener comprehension through contextual cues. For instance, the non-lexicalized sequence “*J’ai l’impression, je crois qu’il*” (*I have the impression, I believe that he..*) is produced as a temporally reduced stretch [ʒɛlɛpɾɛsjɔʒkɾwaki] where 17 phonemes are compressed into a 537ms window¹, making it difficult to isolate as a sequence of discrete units (Figure 1). Such reductions reflect complex articulatory motor patterns where speakers economize effort while still providing sufficient material for listener comprehension.

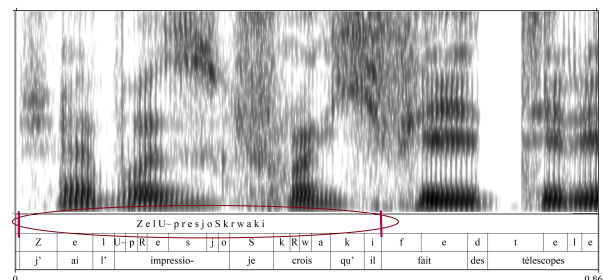


Figure 1: An example of a non-lexicalized reduction: reduced sequence (top tier), segments corresponding to the sequence “*j’ai l’impression je crois qu’il*” [ʒɛlɛpɾɛsjɔʒkɾwaki], (second), and the involved tokens (bottom). The sequence is made up of 17 phonemes produced in 537ms.

This study examines the relationship between vowel space properties and non-lexicalized reductions. Specifically, it explores whether speakers with smaller vowel space areas (measured by pVSA) and lower vowel distinctiveness (measured as VDI) —both indicative of greater reduction— are more prone to produce non-lexicalized reductions characterized by temporally compressed segments, where vowel shortening is hypothesized to cause formant undershoot and reduced vowel spaces

¹This sequence, along with other audio-visual examples of reductions, will be available in a repository upon acceptance.

[4]. Articulation rate, as a potential modulator of both pVSA and VDI, is also considered. By investigating these metrics, this study seeks to clarify the interplay between vowel reduction and non-lexicalized speech patterns, offering insights into how individual variability in vowel systems shapes broader phonetic and linguistic processes. Additionally, the bottom-up approach for detecting perceptually subtle, temporally compressed reductions helps expand the scope of reduction analysis.

2. Method and Materials

2.1. Dataset

The Corpus of Interactional Data² [12] was used to extract vowels and reductions. The dataset includes eight one-hour audio-visual recordings of conversations between same-gender pairs of colleagues (10 females, 6 males; $M = 34$ years), totaling eight hours of interaction. All participants were native French speakers. The corpus includes comprehensive multimodal annotations across linguistic levels, such as gestures, prosody, syntax, and phonetics. The aligner used for segmentation identifies only seven macro-classes of oral vowels, without distinguishing between mid-closed and mid-open variants. These macro-classes are represented as a, e [e, ɛ], o [o, ɔ], ø [ø, œ, ə], u, y, and i,³ reflected in the corpus' phonetic annotations. For this study, we used an improved version (v3) of the phonetic annotations. This version includes better phoneme alignment with the help of SPPAS [13] and additional manual corrections.

2.2. Vowel Extraction

Our analysis focuses exclusively on French oral vowels, excluding nasal vowels. Vowel properties of the CID were extracted using the Praat script described in [9]. For each vowel, the script computed formant values at three points: start, middle, and end. Only the mean formant values across these points were used for the analysis. The script identified 113,296 oral vowels in the corpus. Following the application of a duration filter to retain only vowels with durations between 30 and 400ms (as such durations are unlikely to reflect meaningful linguistic reductions) a subset of 112,346 vowels were selected for analysis, of which 6373 were found in the detected reduced sequences.

Lobanov normalization, performed with the *norm.Lobanov* function from the "vowels" package [14], was applied to control for speaker variability and enable consistent inter-speaker comparisons. This method, chosen for its sensitivity to individual differences, ensures reliable metrics derived from normalized acoustic values.

2.3. Acoustic Metrics

pVSA. Vowel Space Area is a phonetic measure that quantifies the articulatory space occupied by vowels produced by a speaker. It reflects the dispersion and distinctiveness of vowel sounds, providing insights into a speaker's articulatory and phonetic characteristics. Vowel space area might be influenced by factors such as speaker anatomy [15], language and dialect [16], speaking style [17], age, gender [18], and speech rate [19]. The Pentagonal Vowel Space Area (pVSA) was computed based on five corner vowels /a, e, i, u, o/ (after the duration filter, a to-

tal of 93,361 corner vowels were identified in the CID). The detailed formula⁴ used to compute the pVSA can be found in [20]⁵. Smaller pVSA values indicate reduced vowel spaces, which we hypothesize to be correlated with higher rates of non-lexicalized reductions.

$$pVSA = \frac{1}{2} \sum_{i=v} (F1_i F2_{i+1} - F1_{i+1} F2_i)$$

VDI. Vowel distinctiveness index refers to the clarity and separability of vowel sounds, influenced by formant frequencies, vowel duration, and coarticulatory effects. While acoustic cues are essential, speaker-related factors such as dialect, accent and speech rate contribute substantially to distinctiveness. The VDI as described by [22] is based on the ratio of the overall vowel space to the dispersion of each vowel category, where dispersion is measured in terms of mean Euclidean distances between the vowel/system centers and their periphery. It reflects the relationship between vowel space size and category dispersion, with higher values indicating hyper-articulated speech and lower values suggesting hypo-articulation. The VDI scores of speakers were computed following the formula⁶ detailed in [23].

$$\frac{D_{SYS}}{d_{VOY}} = \frac{\sqrt{(F1v - F1S)^2 + (F2v - F2S)^2}}{\sqrt{(F1v - F1V)^2 + (F2v - F2V)^2}}$$

Articulation Rate. Articulation rates, calculated as the number of phonemes produced per second (excluding pauses), were extracted by Chardenon and colleagues [24] for the speakers in CID.

2.4. Detecting Reductions

Temporally and phonemically reduced zones were identified using a bottom-up detection approach. A custom script⁷ detected sequences containing at least 6 phonemes in a 230ms sliding window. Processing phoneme alignment files with temporal boundaries, the script counts phonemes from each onset. Once at least six phonemes are identified within the window, the sequence is recorded by shifting one phoneme, and the script goes on to search for the following 230ms window. If the threshold (6 phonemes) is exceeded, the window expands to include additional phonemes until the phoneme count in the sliding 230ms window falls below the threshold. The 230 ms window was selected after testing smaller sizes (e.g., 180ms, 200ms), which failed to provide a sufficient number of sequences for a reliable analysis.

A total of 956 highly reduced sequences were identified, corresponding to 2.1% of all phonemes in the CID (6,351 phonemes out of 297,703). Speaker reduction ratios (NLR%) were calculated as the proportion of phonemes in reduced sequences relative to the total phonemes produced by each speaker.

⁴The index i represents the position of a vowel in the ordered sequence of vowel categories used to define the vocalic pentagon.

⁵*convexHullArea* and *VowelMeansPolygonArea* functions from "phonR" package [21] were also computed to provide information regarding vowel space area. Both provided compatible results with our pVSA values.

⁶'v' refers to vowels (exemplar), 'V' refers to a vowel category while 'S' refers to the vowel system.

⁷To preserve anonymity, the detection script's source code will be available on GitHub upon acceptance.

²The corpus, along with annotations, is freely accessible under open licenses and is available on the "Ortolang" repository: <https://hdl.handle.net/11403/ortolang-000722/v1>

³Detailed information regarding phonetic annotation and automatic alignment can be found in [12].

Group	Male						Female									
Speaker	AG	AP	EB	LJ	SR	YM	AB	AC	BX	CM	IM	LL	MB	MG	ML	NH
pVSA	2722	5976	1561	4715	3742	1190	6430	2846	3550	2665	6486	5724	7178	9337	4321	7810
VDI	1,15	1,17	0,85	1,11	1,3	0,8	1,22	0,90	0,73	0,90	0,98	0,60	1,07	1,02	0,99	0,78
NLR	94	87	215	45	20	45	41	79	64	46	15	7	47	48	63	41
NLR%	3,7	2,7	5,5	1,3	0,9	1,6	1,7	3,3	2,7	1,6	0,5	0,6	1,4	1,6	2,6	1,6
Art.Rate	13,1	13,26	14,4	11,45	11,19	12,17	11,9	13,08	13,42	12,16	11,87	11,08	12,37	13,18	12,33	13,31

Table 1: Values for individual speakers, including pentagonal vowel space area (pVSA), vowel distinctiveness index (VDI), counts and ratios of non-lexicalized reduction sequences (NLR and NLR%), and articulation rates measured in phonemes per second.

3. Results

This section presents the pVSA and VDI values, followed by an analysis of the interaction between vowel space metrics and speakers' reductions. Individual values of pVSA, VDI and non-lexicalized reductions ratios (NLR%) are reported in Table 1.

The mean pVSA value was 5635 ($SD = 2233$, $range : 2665 - 9337$) for female speakers and 3318 ($SD = 1853$, $range : 1190 - 5976$) for male speakers. The mean VDI was 0.92 ($SD = 0.18$, $range : 0.6 - 1.2$) for female speakers and 1.05 ($SD = 0.2$, $range : 0.8 - 1.3$) for male speakers. While no significant difference was found in VDI values between genders ($Welch's t - test : t(9.66) = -1.30$, $p = 0.222$), pVSA values were significantly higher for female speakers ($t(12.31) = 2.24$, $p = 0.044$).

A Bayesian linear regression model was fitted to investigate the relationship between NLR and two predictor variables pVSA and VDI. The model also included random intercepts based on the variable gender to account for possible inter-speaker variation that might arise from gender differences in reduction patterns ($NLR \sim pVSA + VDI + (1|gender)$). Detailed results are presented in Table 2.

Parameter	Est	SE	95% CI	ESS	Rhat
Intercept	2.08	0.81	[0.26, 3.95]	288	1.02
pVSA	-0.63	0.41	[-1.49, 0.16]	1417	1.00
VDI	0.06	0.39	[-0.71, 0.87]	1428	1.00
$sd(Intercept) (gender)$	1.09	1.09	[0.03, 3.99]	658	1.00
σ	1.33	0.29	[0.89, 2.03]	1403	1.00

Table 2: Summary of Model 1 showing estimates (EST), standard errors (SE), 95% credible intervals, effective sample size (ESS: Bulk), and convergence diagnostic (Rhat). Includes multilevel hyperparameters for gender and residual standard deviation.

The standard deviation of the random intercept for gender was 1.09, indicating moderate variability in NLR between the two gender groups. The fixed-effect intercept, which represents the average NLR when predictors are centered and gender variability is accounted for, was estimated at 2.08 ($Est.Error = 0.81$). The pVSA coefficient showed a negative association with NLR ($Estimate = -0.63$), though the credible interval included zero, suggesting the effect is not statistically significant. The VDI parameter showed a small, positive, but non significant effect ($Estimate = 0.06$). The residual variability (σ) was estimated at 1.33, suggesting additional unexplained variation in NLR. All parameters exhibited good convergence ($\hat{R}$ values close to 1) and sufficient effective sample sizes ($Bulk_ESS$ and $Tail_ESS > 1000$), ensuring robust posterior estimates. These results indicate that the predictors pVSA and VDI did not exhibit strong evidence of NLR.

Given that the primary predictors (pVSA and VDI) did not demonstrate significant effects, a follow-up model was

fitted to further investigate these relationships. This revised model introduced articulation rate as another predictor variable, while keeping the random effect of gender constant ($NLR \sim pVSA + VDI + articulation + (1|gender)$), to explore whether articulation rate adds explanatory power to the frequency of NLR.

In Model 2, articulation as a fixed effect significantly improved the model, showing a positive and significant relationship with NLR ($Estimate = 1.12$, $CI = [0.76, 1.46]$). This suggests that higher articulation rate is strongly linked to increased NLR. A negative association was found between pVSA and NLR (-0.59), suggesting that smaller vowel spaces might correspond to higher NLR values. In contrast, VDI (0.25) had a small and non-significant positive effect, indicating no significant influence on NLR. The Intercept was estimated at -11.86 , indicating a low baseline NLR, but this effect is also accompanied by substantial uncertainty ($95\%CI : [-16.35, -7.53]$). The variability due to gender ($sd(Intercept)$) (0.79) reflected variability between male and female speakers in NLR. The addition of articulation led to a reduction in the random intercept variability for gender, suggesting that articulation may account for some of the variability previously attributed to gender. Additionally, the residual standard deviation (σ) decreases from 1.33 in Model 1 to 0.59 in Model 2 (see Table 3), reflecting an improved fit of the model with articulation included.

Parameter	Est	SE	95% CI	ESS	Rhat
Intercept	-11.86	2.27	[-16.35, -7.53]	1178	1.00
pVSA	-0.59	0.19	[-0.98, -0.24]	577	1.01
VDI	0.25	0.18	[-0.13, 0.58]	1253	1.00
Articulation	1.12	0.17	[0.76, 1.46]	1518	1.01
$sd(Intercept) (gender)$	0.79	0.84	[0.02, 2.89]	328	1.01
σ	0.59	0.15	[0.39, 0.96]	325	1.02

Table 3: Summary of Model 2 showing estimates (EST), standard errors (SE), 95% CI, effective sample size (ESS: Bulk) and convergence diagnostic (Rhat) for the relationship between NLR and predictor variables (pVSA, VDI, articulation).

Finally, a third Bayesian regression model was fitted to incorporate interaction terms between articulation, pVSA, and VDI, to see how articulation might influence the relationship between vowel space area (pVSA) and distinctiveness (VDI) in shaping NLR ($NLR \sim pVSA * articulation + VDI * articulation + (1|gender)$). The articulation variable remained a significant positive predictor of NLR in this model ($Estimate = 1.08$, $95\%CI [0.78, 1.37]$). Interestingly, the pVSA showed a significant positive main effect ($Estimate = 4.72$), but its interaction with articulation (pVSA:articulation) has a small but significant negative effect on NLR (-0.41). This suggests that the effect of pVSA on NLR decreases as articulation rate increases. The main effect of VDI was negative but non-significant ($Estimate = -1.50$), and its interaction with articulation (articulation:VDI) was small and non-

significant ($Estimate = 0.15$, $95\%CI : [-0.13, 0.42]$). This indicates that VDI contributed only minimally to explaining NLR variation. The intercept was estimated at -11.51 ($95\%CI : -15.53, -7.58$), indicating that when all predictors are at their mean, the baseline NLR is negative. The random effect of gender ($sd(Intercept)$) (0.81) indicated some variability across genders, while the residual error ($sigma$) (0.46) suggested good model fit. This value decreased from 0.59 in Model 2 to 0.46 in Model 3, indicating that the addition of interaction terms improved the overall model fit. Overall, this model suggests that articulation is consistently the strongest predictor of NLR. While pVSA has a significant positive effect, its interaction with articulation reduces its impact.

Parameter	Est	SE	95% CI	ESS	Rhat
Intercept	-11.51	1.99	[-15.53, -7.58]	1982	1.00
pVSA	4.72	2.12	[0.41, 8.78]	1641	1.00
Articulation	1.08	0.15	[0.78, 1.37]	2240	1.00
VDI	-1.50	1.64	[-4.83, 1.79]	2033	1.01
pVSA:Articulation	-0.41	0.16	[-0.73, -0.08]	1642	1.00
Articulation:VDI	0.15	0.14	[-0.13, 0.42]	2028	1.01
$sd(Intercept)$ (gender)	0.81	1.00	[0.02, 3.66]	814	1.01
$sigma$	0.46	0.14	[0.28, 0.81]	697	1.00

Table 4: Summary of the Model 3 with articulation as an interaction term, summarizing estimates for fixed effects, interaction terms, and random effects predicting NLR. Includes estimates, standard errors (SE), 95% confidence intervals (CI), effective sample size (ESS: Bulk) and Rhat values.

4. Discussion

This study investigated the relationship between vowel space properties and non-lexicalized reductions, hypothesizing that a reduced vowel system—characterized by a shift from clearer, distinct articulatory patterns towards more centralized and less distinct sounds, reflected in a smaller vowel space—would correspond to higher rates of non-lexicalized reductions. These reductions, identified as temporally compressed sequences of segments within the speech flow, were expected to be more frequent among speakers with smaller, less distinct vowel spaces. This prediction stemmed from the observation that shorter durations—typical of segments in non-lexicalized sequences—tend to result in degraded F1 and F2 values, further contributing to reduced vowel space size and distinctiveness. Additionally, articulation rate was considered as a key factor shaping this relationship, offering a dynamic view of how vowel space properties interact with speech patterns.

To test these hypotheses, the CID dataset was chosen for its familiar and spontaneous conversational context, which is thought to encourage speakers to prioritize communicative efficiency and naturalness over articulatory precision. Such a setting fosters greater speech variability and more frequent instances of reduction compared to read or controlled speech [25]. In this dataset, reduction rates and acoustic metrics revealed notable variability among speakers. Interestingly, some speakers exhibited large vowel systems with low distinctiveness (e.g., NH; $pVSA : 7810$, $VDI : 0, 8$) while others displayed relatively small vowel spaces with high distinctiveness (e.g., SR; $pVSA : 3742$, $VDI : 1, 3$). A post-hoc correlation test found no significant relationship between pVSA and VDI values ($p = 0.48$), indicating that vowel systems can vary in structure independently of overall distinctiveness.

Our findings highlight articulation rate as the strongest predictor of NLR across models, emphasizing the role of speech

tempo in driving reductions. This aligns with the Hyper- and Hypo-speech framework [1], which posits that increased speech rate compromises articulatory precision, leading to formant undershoot and thus reduction in speech. Consistent with [26] and [27], this study underscores the strong relationship between temporal dynamics and vowel reduction. On the other hand, [28] found that while average vowel space size does not significantly differ between high and low speech rates, slow talkers exhibit greater inter-speaker variability and generally larger vowel spaces. This adds to how individual articulatory behavior—beyond articulation rate alone—may contribute to reduction patterns, underscoring the interaction of speaker-specific factors in shaping reduction phenomena.

Regarding pVSA, our findings demonstrate that smaller vowel space areas predict greater NLR, but only when articulation rate is taken into account in the model. This suggests that the impact of vowel space size on NLR is not static; instead, it interacts with speakers’ temporal dynamics to shape reduction patterns. At slower articulation rates, smaller pVSA likely reflects biomechanical constraints, leading to reductions via reduced clarity of vowel articulation. However, at faster rates, reductions are dominated by the temporal compression itself. This finding extends the work of Perkell et al. [29], who argued that vowel space area reflects both cognitive and physical constraints in speech production which are modulated by temporal demands. By showing that pVSA’s significance is contingent on articulation, these results extend prior research by highlighting the interaction between spatial and temporal dimensions of vowel production.

VDI, as a measure of vowel system dispersion relative to vowel category dispersion, did not consistently emerge as a significant predictor of NLR. As a theoretically robust index, VDI reflects the ratio of global vowel system variability to local category variability, providing insight into how distinct vowel categories are relative to the overall system. One explanation for its limited predictive power in this study may relate to the scope of NLR as a phenomenon. While NLR might reflect temporal compression and formant undershoot, VDI’s theoretical underpinnings may be more sensitive to broader systemic patterns of vowel articulation rather than local reductions. Additionally, its relationship with NLR may depend on unmeasured factors such as phonological environment or individual speaking strategies. These results suggest that vowel distinctiveness might interact with other contextual variables to shape reduction.

Notably, gender effects diminished when articulation rate and vowel space measures were included in the model, suggesting that previously observed gender differences in reduction may be better explained by articulatory dynamics rather than inherent gender-based speech patterns. This aligns with previous research [30; 31], which emphasizes articulation rate and biomechanical constraints over categorical gender effects. This study reinforces the central role of articulation rate in non-lexicalized reductions, with vowel space size influencing reduction only when considered alongside speech tempo. These findings highlight the interaction between spatial and temporal aspects of speech rather than a direct effect of vowel distinctiveness on reduction patterns. Future studies should explore these metrics and their relationships with reduction behavior in larger, more diverse datasets to account for variability across languages, dialects, or speech contexts. Additionally, while this study considered articulation rate, other factors—such as cognitive load, task difficulty, and social influences—may also shape vowel reduction patterns and should be incorporated into future models.

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